Honours Linear Algebra - Week 2

14 and 16 September 2026

14 Sept. / Four spaces; independence

$A \in M_{m\times n}(K)$:

subspaces of K^n	subspaces of K^m
$\mathrm{Nul}(A) = \{x : Ax = 0\}$	$\mathrm{Nul}(A^T)$
$\mathrm{Row}(A) = \mathrm{Span}\{\text{rows}\}$	$\mathrm{Col}(A) = \mathrm{Span}\{\text{columns}\}$

$A(au + bv) = aAu + bAv$: null spaces closed. Spans are subspaces (wk1).

$$\mathrm{Row}(UA) \subseteq \mathrm{Row}(A), \qquad \mathrm{Col}(AB) \subseteq \mathrm{Col}(A).$$

Rows/columns of the product are combinations of the originals. If U, B invertible, equality: apply U^{-1}, B^{-1} for the reverse inclusion.

Finitely generated: $V = \mathrm{Span}(v_1, \ldots, v_r)$ for some finite list. K^n, $K_n[t]$: yes. $C^0(\mathbb{R})$: no ($1, x, \ldots, x^N$ independent for every N).

Def.1.14–15.

$$v_1, \ldots, v_r \text{ independent} \iff \left[\sum_i c_i v_i = 0 \Rightarrow c_1 = \cdots = c_r = 0\right].$$

Dependent: a nonzero coefficient list gives zero. Equivalently, some v_k lies in the span of the others:

$$\sum_i c_i v_i = 0, \quad c_k \neq 0 \quad \Rightarrow \quad v_k = -\sum_{i\neq k} \frac{c_i}{c_k} v_i.$$

Singleton: independent iff $v \neq 0$. Pair: dependent iff one is a multiple of the other. Any list containing 0: dependent.

Ex.1.33–35. Explicit relations:

$$\begin{pmatrix}1\\2\\3\end{pmatrix} - 2\begin{pmatrix}4\\5\\6\end{pmatrix} + \begin{pmatrix}7\\8\\9\end{pmatrix} = 0,$$
$$t^2 - 4t + (4t - t^2) = 0,$$
$$\sin 2x - 2\sin x \cos x = 0.$$

Ex.1.36 / field matters. $\{\sin x, \cos x, e^{ix}\}$, as complex-valued functions:

$$e^{ix} - \cos x - i \sin x = 0 \quad \Rightarrow \quad \text{dependent over } \mathbb{C}.$$

Real coefficients: imaginary parts force the e^{ix} coefficient to vanish; evaluate at $0, \pi/2$ for the other two. Independent over $\mathbb{R}$.

Bases; coordinates

Def.1.16. Basis $\mathcal{B} = (b_1, \ldots, b_n)$: independent + spanning. Standard bases: $(e_1, \ldots, e_n)$ for K^n; $(1, t, \ldots, t^n)$ for $K_n[t]$.

Thm.1.17 / remove redundant vectors. For finite S,

$$v_k \in \mathrm{Span}(S \setminus \{v_k\}) \Rightarrow \mathrm{Span}(S) = \mathrm{Span}(S \setminus \{v_k\}).$$

Substitute the expression for v_k into each combination. Repeat until independent $\Rightarrow$ a basis remains. Minimal spanning set = basis; $\{0\}$ has empty basis.

Thm.1.18–19 / unique coordinates.

$$\mathcal{B} \text{ basis} \iff \forall v \in V \ \exists!\, (c_1, \ldots, c_n) : \quad v = \sum_i c_i b_i, \qquad [v]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

Uniqueness: subtract two expressions; independence gives $c_i - c_i' = 0$. Conversely, existence gives spanning; uniqueness for 0 gives independence.

Ex.1.39. $p = t^3 + 2t^2 + 3t + 4$, $\mathcal{E} = (1, t, t^2, t^3)$:

$$[p]_{\mathcal{E}} = (4, 3, 2, 1)^T.$$

Order matters. Invertible $n \times n$ matrix: columns form a basis of K^n, since $Ax = b$ has a unique solution for every b.

Computing bases. Row-reduce $A \to R$.

- $\mathrm{Row}(A)$: nonzero rows of R (independent by successive pivot columns).
- $\mathrm{Col}(A)$: pivot columns of the **original** A; alternatively row-reduce A^T.
- $\mathrm{Nul}(A)$: solve $Rx = 0$, one vector per free parameter.

Row operations preserve relations among columns: $Ac = 0 \iff Rc = 0$. They need not preserve the column space itself.

Ex. / null basis.

$$A = \begin{pmatrix} 1 & -1 & 4 & 5 & 4 \\ 0 & 0 & 1 & 2 & 6 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \begin{cases} x_5 = 0, \\ x_3 = -2t, \\ x_2 = s, \ x_4 = t, \\ x_1 = s + 3t. \end{cases}$$

$$x = s \underbrace{\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}}_{v_1} + t \underbrace{\begin{pmatrix} 3 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix}}_{v_2}.$$

$\{v_1, v_2\}$ spans the null space; coordinates $2, 4$ force independence. Hence a basis.

16 Sept. / Dimension

Thm.1.20 / replacement. $V = \mathrm{Span}(v_1, \ldots, v_n)$ and $u_1, \ldots, u_m$ independent $\Rightarrow m \le n$.

Proof via equations: write $u_i = \sum_j a_{ji} v_j$. If $m > n$,

$$\sum_{i=1}^{m} a_{ji} x_i = 0 \quad (j = 1, \ldots, n) \Rightarrow \exists x \neq 0 \Rightarrow \sum_{i=1}^{m} x_i u_i = 0,$$

contradiction. n homogeneous equations, $m > n$ unknowns.

Thm.1.21–22. Two finite bases: apply replacement both ways $\Rightarrow$ same size. Define

$$\dim_K V = |\mathcal{B}|, \qquad \dim\{0\} = 0.$$

Not finitely generated: $\dim V = \infty$ (lecture convention).

V	$\dim_K V$	V	$\dim_K V$
K^n	n	$K_n[t]$	$n+1$
$M_{m\times n}(K)$	mn	$K[t]$	∞

$x + y + z = 0$ in $\mathbb{R}^3$: basis $(1, -1, 0)^T, (1, 0, -1)^T$, dimension 2. $\dim_{\mathbb{R}} \mathbb{C} = 2$, $\dim_{\mathbb{C}} \mathbb{C} = 1$; $\dim_{\mathbb{Q}} \mathbb{R} = \infty$.

Thm.1.23 / extend independent vectors. $\dim V = n < \infty$, $U \subseteq V$ a subspace. Any independent list in U extends to a basis of U; $\dim U \le n$.

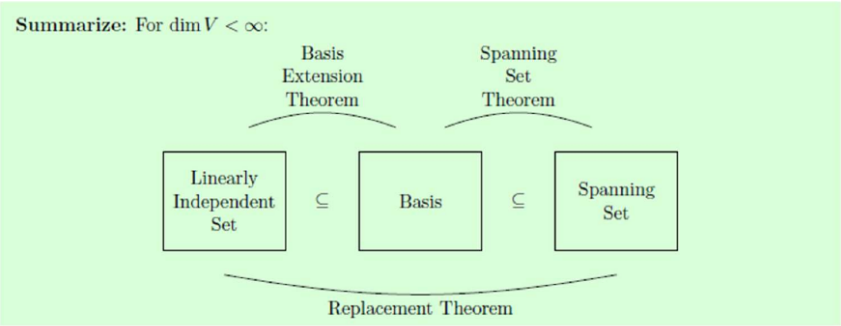
If not spanning, choose $u \in U \setminus \mathrm{Span}(S)$. Then $S \cup \{u\}$ stays independent. Repeat; replacement stops the process by n vectors.

Cor.1.24. $U \subseteq V$, finite dimensions equal $\Rightarrow U = V$. Otherwise extend a basis of U by one more vector in V.

Thm.1.25. If $|S| = \dim V = n$, then

$$S \text{ independent} \iff S \text{ basis} \iff S \text{ spanning}.$$

Extend / delete; no room to add / remove. The count n is essential.



16 Sept., p.11. Extend independent sets; trim spanning sets.

$\mathbb{R}^3$ subspaces: $\{0\}$, lines through 0, planes through 0, $\mathbb{R}^3$.

Rank; turning a row space into equations

Definitions. Row rank $= \dim \mathrm{Row}(A)$; column rank $= \dim \mathrm{Col}(A)$; nullity $= \dim \mathrm{Nul}(A)$.

With r pivots, n columns:

$$\dim \mathrm{Row}(A) = \dim \mathrm{Col}(A) = r, \qquad \dim \mathrm{Nul}(A) = n - r.$$

Nonzero echelon rows / original pivot columns / free-parameter vectors give the respective bases.

Ex. / 16 Sept., pp.12–13. For A as printed:

$$A = \begin{pmatrix} 1 & -1 & 1 & 5 & 4 \\ 1 & -1 & 5 & 7 & 10 \\ 2 & -2 & 8 & 10 & 9 \\ 3 & -3 & 13 & 17 & 19 \\ 0 & 0 & 2 & 4 & 13 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Four pivots, not three. The slide's echelon matrix and null basis do not match this A.

$$\begin{aligned} \mathcal{B}_{\mathrm{row}} &= \{(1,-1,0,0,0),\ e_3^T, e_4^T, e_5^T\}, \\ \mathcal{B}_{\mathrm{null}} &= \{(1,1,0,0,0)^T\}, \\ \mathcal{B}_{\mathrm{col}} &= \{A_{\cdot 1}, A_{\cdot 3}, A_{\cdot 4}, A_{\cdot 5}\}. \end{aligned}$$

Also $\mathrm{Nul}(A^T) = \mathrm{Span}\{(0,-1,-1,1,0)^T\}$: row 4 = row 2+ row 3.

Find B with $\mathrm{Nul}(B) = \mathrm{Row}(A)$. Take the null-basis vector as a row:

$$B = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \end{pmatrix}, \qquad BA^T = 0.$$

Hence $\mathrm{Row}(A) \subseteq \mathrm{Nul}(B)$, and

$$\dim \mathrm{Row}(A) = 4 = 5 - 1 = \dim \mathrm{Nul}(B) \Rightarrow \mathrm{Row}(A) = \mathrm{Nul}(B).$$

Here both are simply $\{y \in \mathbb{R}^5 : y_1 + y_2 = 0\}$.

Direct sums

Def.1.26–28. U, W subspaces of V:

$$U + W = \{u + w : u \in U, w \in W\} = \mathrm{Span}(U \cup W).$$

$U \cap W$ and $U + W$ are subspaces (closure). Direct sum means

$$V = U \oplus W \quad \Longleftrightarrow \quad V = U + W, \quad U \cap W = \{0\}.$$

Thm.1.29. Direct sum $\Longleftrightarrow$ every $v \in V$ has a unique decomposition $v = u + w$.

$$u + w = u' + w' \Rightarrow u - u' = w' - w \in U \cap W = \{0\}.$$

Conversely, $z \in U \cap W \Rightarrow z = z + 0 = 0 + z \Rightarrow z = 0$ by uniqueness.

Ex.1.48–51.

$$\begin{aligned}
\mathbb{R}^3 &= \mathrm{Span}(e_1, e_2) \oplus \mathrm{Span}(e_3), \\
K[t] &= K \cdot 1 \oplus \{p : p(0) = 0\}, \qquad\qquad p(t) = p(0) + [p(t) - p(0)], \\
f(x) &= \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd}}, \\
A &= \underbrace{\frac{A + A^T}{2}}_{\text{symmetric}} + \underbrace{\frac{A - A^T}{2}}_{\text{skew-symmetric}}.
\end{aligned}$$

Last two over $\mathbb{R}$ (more generally, characteristic $\neq 2$). Intersections zero: even + odd forces $f = -f$; symmetric + skew forces $A = -A$.

Thm.1.30 / complement. $\dim V < \infty$, $U \subseteq V$ a subspace $\Rightarrow \exists W$ with $V = U \oplus W$.

$$\underbrace{(u_1, \ldots, u_r)}_{\text{basis of } U} \longrightarrow \underbrace{(u_1, \ldots, u_r, w_1, \ldots, w_s)}_{\text{basis of } V}, \qquad W = \mathrm{Span}(w_1, \ldots, w_s).$$

Spanning gives $V = U + W$; independence gives $U \cap W = \{0\}$. Complement not unique:

$$\mathbb{R}^2 = \mathrm{Span}(e_1) \oplus \mathrm{Span}(e_2) = \mathrm{Span}(e_1) \oplus \mathrm{Span}(e_1 + e_2).$$

Infinite-dimensional versions: existence of bases / complements uses the Axiom of Choice.